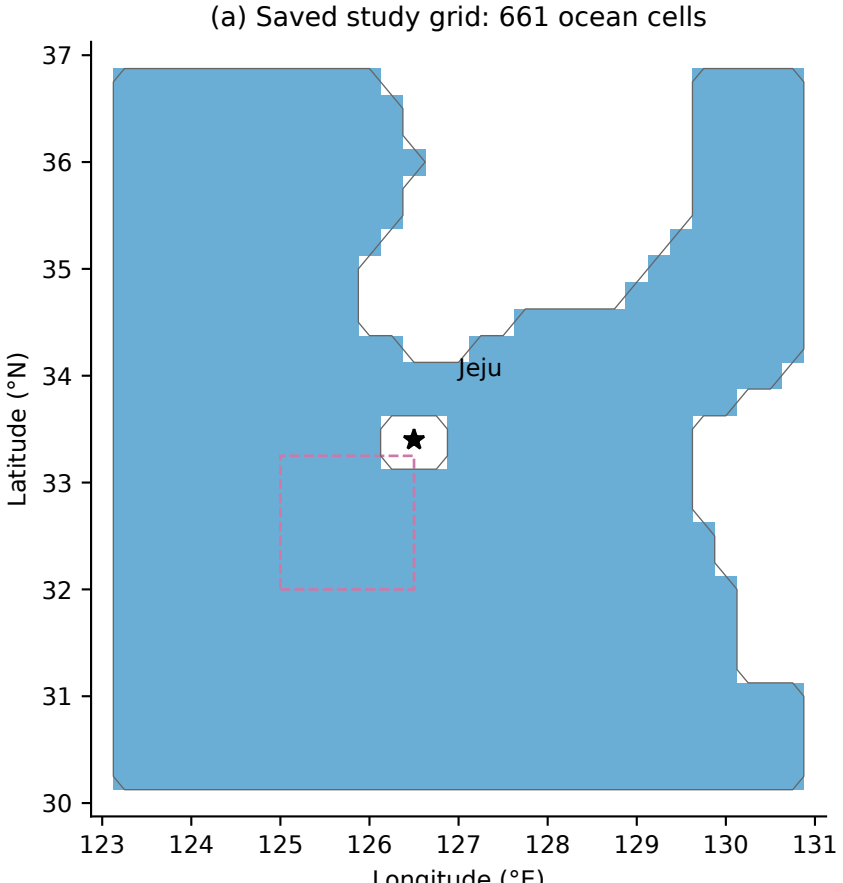

Data and split

1993–2022 daily SST + physical fields
30-day history; fixed Validation and Test periods

Nested training subsets

10%: 376 ⊂ 25%: 987 ⊂ 50%: 1,974 ⊂ 100%: 3,995 windows
90-day blocks; subset-specific training statistics

Five matched models

Baseline | Auxiliary-only | P0 | NoAdv | Adv
Same architecture; identical ratio / lead / seed pairing

Training and evaluation

2 leads (7, 14 days) × 3 seeds × 4 ratios × 5 models = 120 runs
80 actual epochs; Validation-best; exploratory Test
Oracle physical forcing used in training constraint only